BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

1. Introduction

Language model pre-training has been shown to be effective for improving many natural language processing tasks. There are two existing strategies for applying pre-trained language representations to downstream tasks: feature-based and fine-tuning. In this paper, we argue that current techniques restrict the power of pre-trained representations because standard language models are unidirectional, which limits the choice of architectures during pre-training.

BERT Pre-training Objectives

Section 3: Masked Language Model and Next Sentence Prediction

3.1 Masked LM (MLM)

In order to train a deep bidirectional representation, we simply mask some percentage of the input tokens at random, and then predict those masked tokens. We refer to this procedure as a Masked LM (MLM). In all of our experiments, we mask 15% of all WordPiece tokens in each sequence at random. Of the selected tokens, 80% are replaced with the [MASK] token, 10% are replaced with a random token, and 10% are kept unchanged.

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Loss_MLM = - sum_{i in Masked} log P(w_i | Context_{-i})
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3.2 Next Sentence Prediction (NSP)

In order to train a model that understands sentence relationships, we pre-train for a binarized next sentence prediction task that can be trivially generated from any monolingual corpus. When choosing sentences A and B, 50% of the time B is the actual next sentence (labeled as IsNext), and 50% of the time it is a random sentence from the corpus (labeled as NotNext).

Model Sizes, GLUE Benchmark & Limitations

Section 5: Empirical Results across GLUE Tasks

5.1 BERT Architecture Configurations and GLUE Scores

Table comparing BERT architectures and GLUE average accuracy scores against prior baselines:

Model	Layers (L)	Hidden (H)	Heads (A)	Total Params	GLUE Score
OpenAI GPT	12	768	12	117M	72.8%
BERT_BASE	12	768	12	110M	79.6%
BERT_LARGE	24	1024	16	340M	80.5%

6. Computational Costs and Limitations

Pre-training BERT is computationally expensive: BERT_BASE required 4 days on 4 Cloud TPU v2 pods (16 TPU chips), while BERT_LARGE required 4 days on 16 Cloud TPU v2 pods (64 TPU chips). Furthermore, BERT enforces a rigid maximum sequence length limit of 512 tokens due to learned positional embeddings and quadratic self-attention memory requirements.